



Q no. 1

(a) Under what conditions $AB = BA$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix} = \begin{bmatrix} b_{11}a_{11} + b_{12}a_{21} & b_{11}a_{12} + b_{12}a_{22} \\ b_{21}a_{11} + b_{22}a_{21} & b_{21}a_{12} + b_{22}a_{22} \end{bmatrix}$$

Comparing corresponding entries

$$\begin{aligned} a_{11}b_{11} + a_{12}b_{21} &= b_{11}a_{11} + b_{12}a_{21} \\ a_{11}b_{12} + a_{12}b_{22} &= b_{11}a_{12} + b_{12}a_{22} \\ a_{21}b_{11} + a_{22}b_{21} &= b_{21}a_{11} + b_{22}a_{21} \\ a_{21}b_{12} + a_{22}b_{22} &= b_{21}a_{12} + b_{22}a_{22} \end{aligned}$$

$$\begin{aligned} a_{12}b_{21} &= b_{12}a_{21} \\ a_{11}b_{12} + a_{12}b_{22} &= b_{11}a_{12} + b_{12}a_{22} \\ a_{21}b_{11} + a_{22}b_{21} &= b_{21}a_{11} + b_{22}a_{21} \\ a_{21}b_{12} &= b_{21}a_{12} \end{aligned}$$

(b) $A^{r+s} = A^r A^s$ is valid for negative integers r, s

Solution: Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and let $r = 1$ and $s = -1$. It shows, it would be

$$\begin{aligned} A^{1-1} &= A^1 A^{-1} \\ A^0 &= AA^{-1} \\ I &= \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}^{-1} \end{aligned}$$

Inverse not exist, hence it is not valid for negative integers.

(c) Solution: Since A^{-1} does not exist, that's why $AB = AC$ but $B \neq C$. (Fill the details for the verification)

Q.no 2

Solution: Using $[A|I] \rightarrow [I|A^{-1}]$

$$\begin{aligned}
 & \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array} \right] \\
 R_2 - 2R_1 & \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array} \right] & R_3 - 4R_1 & \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \end{array} \right] \\
 R_3 \rightleftharpoons R_2 & \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & -1 & -1 & -2 & 1 & 0 \end{array} \right] & R_3 + R_2 & \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & -1 & -6 & 1 & 1 \end{array} \right] \\
 R_1 + 2R_3 & \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 11 & 2 & 2 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & -1 & -6 & 1 & 1 \end{array} \right] & -1R_3 & \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 11 & 2 & 2 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array} \right] \\
 & \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array} \right] \dots \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -16 & -11 & 3 \\ 0 & 1 & 0 & 7/2 & 5/2 & -1/2 \\ 0 & 0 & 1 & -5/2 & -3/2 & 1/2 \end{array} \right]
 \end{aligned}$$

Q. no. 3

$$x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array} \right]$$

$$\begin{aligned}
 R_2 - 2R_1 & \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & 17 \\ 3 & 6 & -5 & 0 \end{array} \right] & R_3 - 3R_1 & \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array} \right] \\
 R_3 - 3/2R_2 & \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 0 & -1/2 & -3/2 \end{array} \right]
 \end{aligned}$$

$$-1/2x_3 = -3/2 \Rightarrow x_3 = 3$$

$$2x_2 - 7x_3 = -17, \Rightarrow x_2 = 2$$

$$x_1 + x_2 + 2x_3 = 9 \Rightarrow x_1 = 1$$

Question 5

Reduce

$$\left[\begin{array}{ccc} 2 & 1 & 3 \\ 0 & -2 & 7 \\ 3 & 4 & 5 \end{array} \right]$$

to reduced row-echelon form.

Solution:

$$\begin{aligned}
& \begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & 7 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & -2 & 7 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{R_3-3R_1} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & -2 & -29 \\ 0 & 5/2 & 1/2 \end{bmatrix} \\
& \xrightarrow{-\frac{1}{2}R_2} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & 1 & 29/2 \\ 0 & 5/2 & 1/2 \end{bmatrix} \xrightarrow{R_3-\frac{5}{2}R_2} \begin{bmatrix} 1 & 1/2 & 3/2 \\ 0 & 1 & 29/2 \\ 0 & 0 & -143/4 \end{bmatrix} \xrightarrow{R_1-\frac{1}{2}R_2} \begin{bmatrix} 1 & 0 & -13 \\ 0 & 1 & 29/2 \\ 0 & 0 & -143/4 \end{bmatrix} \\
& \xrightarrow{-\frac{4}{143}R_3} \begin{bmatrix} 1 & 0 & -13 \\ 0 & 1 & 29/2 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2-\frac{29}{2}R_3} \begin{bmatrix} 1 & 0 & -13 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1-(-13)R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
\end{aligned}$$

Question 6

Find two different row-echelon forms of

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

Solution: One possibility is

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

Another possibility is

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \xrightarrow{R_1 \rightleftharpoons R_2} \begin{bmatrix} 2 & 7 \\ 1 & 3 \end{bmatrix} \xrightarrow{R_1-\frac{1}{2}R_2} \begin{bmatrix} 1 & 7/2 \\ 1 & 3 \end{bmatrix} \xrightarrow{R_1-R_2} \begin{bmatrix} 1 & 7/2 \\ 0 & 1/2 \end{bmatrix}$$

Question 7 = Ex: 1.3 Q-25

Prove: If A and B are $n \times n$ matrices, then $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$.

Proof. Note that the (i, i) -th entry of $A+B$ is $a_{ii} + b_{ii}$, the (i, i) -th entry of A^T is a_{ii} , and the (i, i) -th entry of B^T is b_{ii} . Then consider the following:

$$\begin{aligned}
\text{Tr}(A+B) &= (a_{11} + b_{11}) + \dots + (a_{nn} + b_{nn}) \\
&= (a_{11} + \dots + a_{nn}) + (b_{11} + \dots + b_{nn}) \\
&= \text{Tr}(A) + \text{Tr}(B)
\end{aligned}$$

Question 8 = Ex 1.3 Q-19

Let A be any $m \times n$ matrix and let 0 be the $m \times n$ matrix each of whose entries is zero. Show that if $kA = 0$, then $k = 0$ or $A = 0$

Proof. Given $kA = 0$ and suppose $k \neq 0$ so,

$$\begin{aligned}
kA &= 0 \\
\frac{1}{k}kA &= \frac{1}{k}0 \\
A &= 0.
\end{aligned}$$

Now suppose $A \neq 0$, if $kA = 0$ and A is nonzero then the only possibility this outcome ($kA = 0$) is $k = 0$.